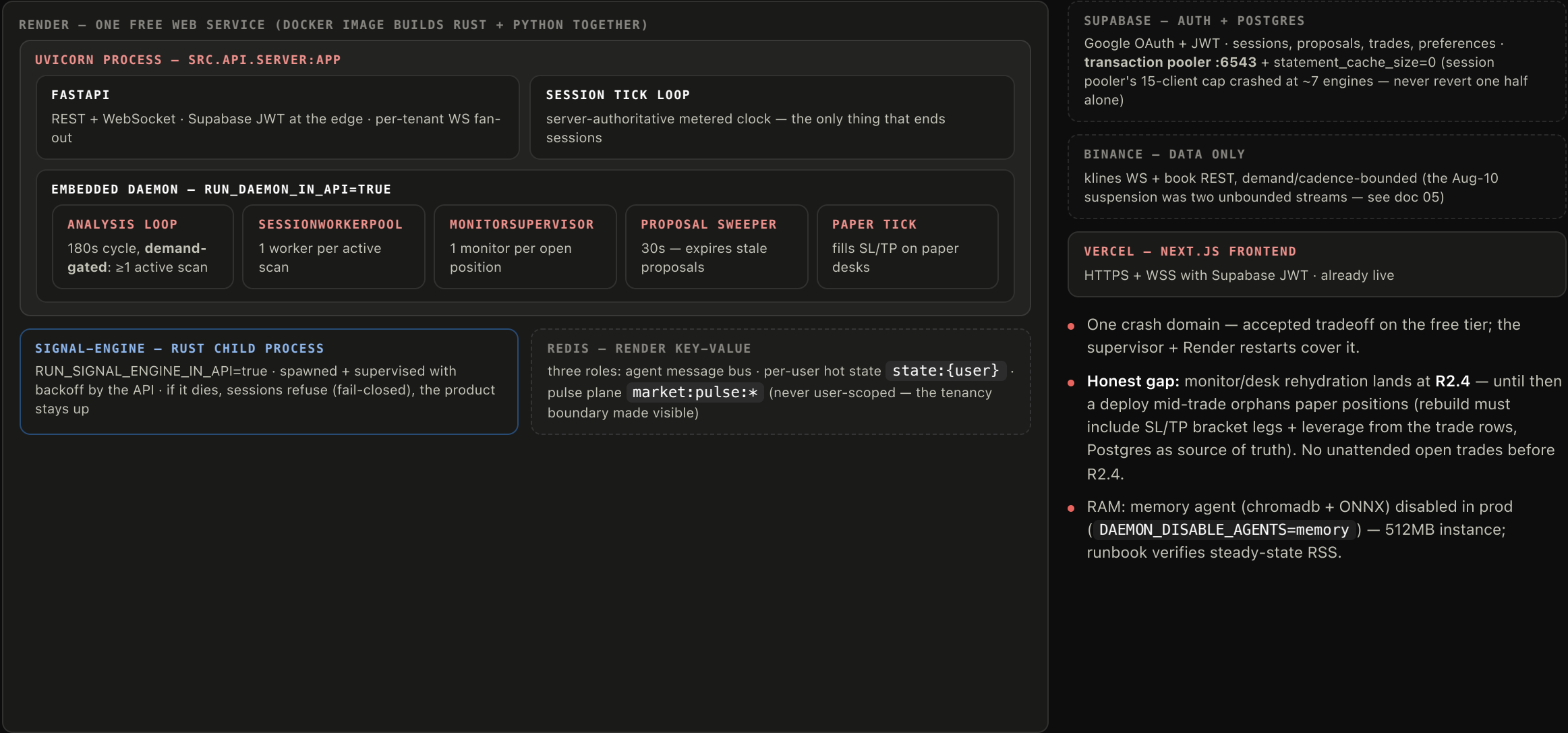


System *design*

The architecture that already runs — one process, three planes, demand-gated compute — plus the exact deltas that turn it into the 2-slot free tier: a slot quota, a trade-focus lock, a hot-window surface, and zero always-on waste.

Doc	Date	Builds on
03 of 06 — System design	2026-08-24	docs/ARCHITECTURE.md (verified 2026-08-11)

One process is the whole backend — by design, for the free tier



Shared facts, per-user judgment, unmetered protection

PLANE 1 · SHARED – USER-BLIND, O(1) IN USERS

Rust engine: ingest (klines WS · book REST) → feature store → 5 vetoes → 6 weighted factors → **regime + tradability 0–100** → **market:pulse:SYMBOL**, refreshed ≤5s, TTL 120s.

SHARED LLM ANALYSIS

Data → Analysis → Strategy agents produce a **candidate-setup pool per symbol** (SharedSetupCache) — computed once per cycle for everyone, demand-gated.

NEW · HOT-WINDOW SURFACE

market:pulse:global — cross-symbol "hot / quiet" now-cast + slots-aware nudge. Every pulse also appends to **pulse_snapshots** so the score stays falsifiable.

The pulse carries facts and scores only — never a direction, entry, or stop. Regime-direction was refuted out-of-sample; a contract test enforces the ban.

PLANE 2 · PER-USER – O(N) IN ACTIVE SCANS

SessionWorker (metered, 1 per scan): reads prefs + fresh pulses — **fail-closed at 15s staleness** — then **analyze_fn**: pulse gate → style filter → **per-user synthesis** (LLM selects the candidate that fits *this* trader and writes the thesis in their terms — it never invents a price) → deterministic sizing vs the user's own risk rules.

PROPOSAL (POSTGRES)

entry / stop / targets / size / leverage / R:R / rationale / shelf life. Clock **pauses** while the user decides. Approval **re-validates at the live price** — drift >0.5% refuses rather than fills worse than agreed.

Per-user LLM budget: **analyze_fn** returns real usage; the cap guards a moving number, fail-closed.

FIX BEFORE ENABLING THE PLANE

Confirmed bug: the pipeline calls **.get()** on **Pulse** dataclasses — enabling **SIGNAL_PLANE_ENABLED** today fails **every metered cycle** while the user's clock drains. Normalize at the boundary (as **monitor_worker** already does) + fold pulse liveness into the capacity check so dead-plane time is refunded.

PLANE 3 · EXECUTION + WATCH

Per-user engine (paper or live) books the bracket. Two money gates fail closed: **ENABLE_EXECUTION** + **LIVE_TRADING_CONFIRMED**.

POSITIONMONITOR – UNMETERED

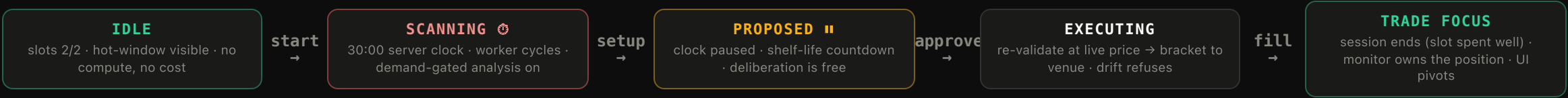
1 per open position. Time stop · profit give-back · **conditions-turned** (pulse drift) → reduce-only exit when the thesis breaks. Outlives the session, restarts, deploys; supervisor reconciles on boot.

BOOKKEEPING

Fill → trade entry row (memory loop feeds) → portfolio + history update → session slot state advances (free tier: focus lock until flat).

The session state machine, extended for slots and trade focus

The metered state machine exists and is well-tested. Two mechanics change for the new product: quota becomes **2 discrete slots/day** (not a minutes pool), and a **focus lock** holds new slots while a free-tier position is open.



SLOT ACCOUNTING (V2 – RED-TEAM HARDENED)

- Today: one pooled 1800s/day quota for every tier, unlimited session count. Replaced by per-plan policy — and **a slot is spent on proposal delivery, not scan start**: a bounded scan (~10 min) with no qualifying setup auto-ends and returns the slot (scan attempts cap 3/day).
- **Slot-native refunds**: a CAPACITY_LOST session re-opens its own slot with the *remaining* grant — never a fresh 30 min (flap test: capacity down/up ×5 must never mint extra scan time).
- **Race-safe consumption**: partial unique index (one active session per user) + slot count and focus-lock check in one transaction; usage is **derived** (COUNT per trading day) — no stored counter, no reset job. Boundary stored UTC, rendered local.
- The sweeper ends any SCANNING session past its grant, **connection or not** — ticks must not depend on an open WebSocket.

THE FOCUS LOCK (FREE TIER)

- Invariant: `open_positions ≥ 1 ⇒ session_start refused`, with the 409 distinguishing **monitored** from "position exists, monitor reattaching".
- **Source of truth: reconciled Redis + engine**, with zombie auto-drop in the lock check — which is why restart-safe rehydration (R2.4) ships **before** the lock (R2.2). `/api/positions/close` stays the escape hatch.
- A slot held by the lock survives exactly **one** midnight reset (intraday holds span midnight routinely).
- Paid tiers raise the concurrency cap; at the cap an async session **pauses unmetered** (new held-at-cap state, mirroring the proposal pause) — skipping synthesis alone would still burn the user's clock.

REJECT / EXPIRE / END PATHS

- Reject or shelf-life expiry → back to SCANNING, clock resumes — covered today.
- Approve → the session **ends** as `trade_opened` (slot spent well). "Trade focus" is a **derived UI state** — open position + monitor status, no session row and no session clock.
- Quota spent or user ends → ENDED; open positions keep their monitors — quota never stops a monitor.

Demand-gated everything: idle costs zero, sessions cost minutes

The founder's instinct becomes the architecture's law: no stream or loop runs without a consumer. The audit enumerated all 25 loops (doc 01); the "always-running 300s loop" (EventCoordinator) was dead code — already attic'd. One blocker remains in live code.

WORKLOAD	CADENCE	TARGET BEHAVIOR	BANDWIDTH (EXTERNAL)	STATUS
Daemon DataAgent streams	demand	DONE · 2896402 — demand = scans OR open exposure (never scans-only: paper fills + monitors read stream-fed prices): scan → klines+markprice; positions → one kline_1m per held symbol; idle+flat → zero streams ; 120s linger; depth/ticker replaced by REST snapshots	~2.2 GB/day → ~0 idle	SHIPPED
Signal engine ingest (child process)	5s score	REST-poll closed bars per TF boundary + 5s batched book poll (the poll is the staleness heartbeat — never stretch it past ~10s to save bytes)	~7.4 GB/mo as WS → ~25 MB/day as REST	R1 · KLINES→REST
Shared LLM analysis cycle	180s	≥1 active scan or explicit keep-warm — zero scans = zero LLM calls (uncommitted diff completes this)	LLM only	DEMAND-GATED
SessionWorkers · PositionMonitors	180s / 15s	1 per active session · 1 per open position; none exist otherwise	internal	DEMAND-GATED
Dashboard market feed	stream	refcounted to live WS viewers + 30s linger (commit 9edada0) — but client-requested extra channels are add-only: refcount them too	~9 MB per viewed hour	FIXED · +R0 REFCOUNT
Idle Binance socket	~90s	skip staleness check while zero streams subscribed — today it reconnect-churns ~960×/day	~6 MB/day	R0 · SMALL FIX
Sweeper · session tick · paper tick · heartheats	30s/5s/3s	always-on but internal-only (Redis/Postgres); idle no-ops against empty sets	0	KEEP
Keepalive cron (GitHub Actions)	10 min	the one deliberate always-on — keeps Render	~0.2 MB/day	KEEP

Schema deltas: small, additive, forward-only

CHANGES FOR THE SLOT MODEL + TIERS	
DELTA	DETAIL
tier config	plans move from env vars to a <code>plans</code> config (free / lite / pro / pro_max): slots_per_day, slot_minutes, max_concurrent_trades, channels, async flag — free tier hardcoded live, others dormant
sessions	slot usage derived — COUNT per trading day (index exists); partial unique index enforcing one active session per user; no stored counter, no reset job (derive-don't-store, the module's own law)
monitor_events	new append-only table (user-indexed, trade-keyed): every monitor verdict + reason. The trade-focus verdict log and exit recap have no backing store today — the monitor publishes latest-only TTL'd Redis status. Prerequisite of the trade-focus UI; survives restarts
user_preferences	capital + risk fields exist — onboarding finally writes them; monthly risk budget (max drawdown %) replaces the return-expectation goal; add <code>onboarding_completed_step</code>
pulse_snapshots	gains its writer (engine appends every publish) — the calibration log the quant desk requires
trades	entry rows written on every book (memory loop) — exit updates must never hit "not found"

- DEBTS ACKNOWLEDGED, SCHEDULED, NOT BLOCKING FREE LAUNCH
- **Money as floats** everywhere except `llm_cost_micros` — one wholesale integer-minor-units + currency migration lands **before any real-money path opens** (roadmap gate, doc 06). Paper-only launch tolerates floats; real money never will.
 - **RLS defense-in-depth** on the five new per-user tables — backend bypasses RLS, so app-layer `user_id` filters remain the only guard; RLS is the second lock.
 - **Alembic vs checkfirst collision** — stamp the production DB once, then forward-only migrations in the deploy step.
- TENANCY INVARIANTS (TESTED, KEEP ENFORCING)
- Every per-user table carries an indexed `user_id`; `pulse_snapshots` must never gain one — schema-parity test enforces both.
 - Every new query filters `user_id` in app code; a missing filter is a launch blocker, not a bug.
 - Per-user Redis keys stay namespaced `state:{user_id}:*`; pulse keys stay global.

API deltas: four new reads, one changed start, nothing else moves

NEW / CHANGED ENDPOINTS	
ENDPOINT	CHANGE
GET /api/pulse	new — per-symbol regime + tradability + global hot/quiet, authenticated, cached 5s server-side; the dashboard's banner + strip read this
GET /api/session/slots	new — slots used/remaining today, focus-lock state + reason, next reset time
POST /api/session/start	changed — no client <code>quota_seconds</code> ; consumes a slot server-side; 409 with monitor status when focus-locked; 503 (no row, no clock) when analysis capacity is down
GET /api/monitors/:symbol/events	new — the monitor's verdict log, read from the append-only <code>monitor_events</code> table (the live status chip keeps using <code>GET /api/monitors</code>)
GET /api/session/recap/:id	new — setups seen / action taken / outcome, powers the recap card

- HARDENING THAT RIDES ALONG (FROM THE AUDIT, DOC 01)

 - `require_admin` fail-closed; anonymous fail-open mode impossible in prod config
 - WS auth moves from `?token=` query param to first-message auth (JWTs out of proxy logs)
 - The static `NEXT_PUBLIC_API_TOKEN` tier dies — Supabase JWT is the only client credential
 - Destructive one-off scripts leave the image (attic'd + .dockerignore)
- CONTRACT RULES

 - Every tenant endpoint requires the JWT and scopes by `user_id` — `/api/setups` 's unauthenticated broadcast model is retired with the v2 daemon path.
 - Pulse responses carry `as_of` + schema version; consumers reject stale or unknown-major — same contract the workers obey.
 - Errors are honest: capacity down = 503 + "engine unavailable", never a silent empty list.